Rational Functions

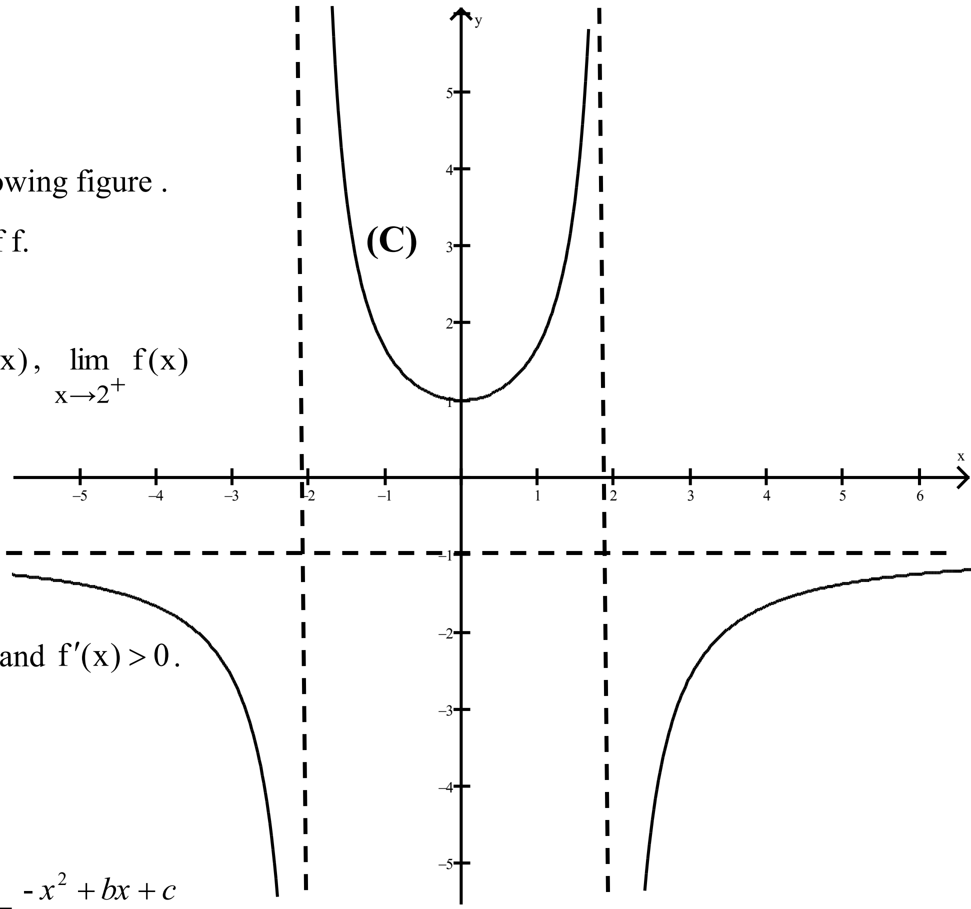
Exercise 1:

Study the variation and draw the curve (C) of the function f defined by: $f(x) = \frac{-2x^2+3x}{x-1}$.

Exercise 2:

Part A

Consider the function f defined by its representative curve (C) given in the following figure .



1) Determine the domain of definition of f.
2) Determine graphically:

   $\lim\limits_{x\to(-2)^-} f(x)$, $\lim\limits_{x\to(-2)^+} f(x)$, $\lim\limits_{x\to2^-} f(x)$, $\lim\limits_{x\to2^+} f(x)$

   $\lim\limits_{x\to-\infty} f(x)$ et $\lim\limits_{x\to+\infty} f(x)$.

   Give the equations of three asymptotes to (C).
3) Solve graphically: $f(x) < 0$, $f'(x) = 0$ and $f'(x) > 0$.
4) Give an equation of an horizontal tangent to (C).
5) Draw the table of variation of f

Part B: The function f is defined by $f(x) = \dfrac{-x^2+bx+c}{x^2-4}$.

1) Determine graphically f (0).
2) Calculate f (0) using the expression f(x).Deduce that c =-4
3) a) Prove that $f'(x) = \dfrac{-bx^2+16x-4b}{(x^2-4)^2}$ then calculate f′(0) in terms of b.

   b) Determine graphically f′ (0). Deduce that b = 0.
4) Suppose that : $f(x) = \dfrac{-x^2-4}{x^2-4}$.

   Write the equation of the tangent to **(C)** at the point of abscissa 1.

Exercise 3:

Let f be the function defined by: $f(x) = \frac{x^2+4x+4}{x+3}$

Designate by (C) its representative curve in an orthonormal $(O, \vec{i}, \vec{j})$

1) Determine the domain of definition of *f.*
2) Calculate the limits on the open bounds of, f. Deduce an asymptote to (C)
3) Verify that: $f'(x) = \frac{x^2+6x+8}{(x+3)^2}$.
4) a) Verify that: $f(x) = x + 1 + \frac{1}{x+3}$.

   b) Prove that (d): y =x+1 is an asymptote to (C).

   c) Study according to the values of x the relative position of (C) and (d).
5) Prove that I (-3 ;-2) is a center of symmetry to (C).
6) Draw the table of variation of *f.*
7) Write the equation of the tangent (T) to (C) at the point of abscissa 1.
8) Trace (C) and (d) in the system $(O, \vec{i}, \vec{j})$.

Exercise 4:

Given the table of variation of a function f and designate by (C) the representative curve of f in an orthonormal system $(O, \vec{i}, \vec{j})$.

x	$-\infty$		-1		1		3		$+\infty$
$f'(x)$		$+$	0	$-$	‖	$-$	0	$+$	
$f(x)$	2	↗	3	↘	$-\infty$ ‖ $+\infty$	↘	1	↗	2

1° Plot (C) knowing that $f(0) = 2$.

2° In the following table, one of the given answers to each question is correct.

Write the number of each question and give the corresponding answer by justifying it.

N°	**Questions**	**Answers**			
		a	**b**	**c**	**d**
1	the domain of definition of f is :	$\mathbb{R}$	$]-\infty ; 1[\cup]1 ; +\infty[$	$]1 ; +\infty[$	$]-\infty ; -1[\cup]3 ; +\infty[$
2	The number of solutions of the equation $f(x) = 0$ is :	2	1	3	0
3	The number of asymptotes to (C) is :	2	1	0	3
4	The number of solutions of the equation $f(x) = 2$ is :	0	3	2	1
5	An equation of the tangent to (C) at A $(-1 ; 3)$ is :	$y = 3$	$y = -x + 3$	$x = -1$	$y = 3x - 1$

Exercise 5:

The adjacent table is the table of variation of a function f defined over $\mathbb{R} - \{0\}$ by $f(x) = -x + a - \dfrac{4}{x}$. ($a$ is a real constant).

x	$-\infty$		-2		0		2		$+\infty$
$f'(x)$		$-$	0	$+$	‖	$+$	0	$-$	
$f(x)$					‖				

1° Find $\lim\limits_{x \to +\infty} f(x)$; $\lim\limits_{x \to -\infty} f(x)$; $\lim\limits_{\substack{x \to 0 \\ x > 0}} f(x)$ and $\lim\limits_{\substack{x \to 0 \\ x < 0}} f(x)$.

2° Complete this table of variation (you are not asked to calculate $f(2)$ and $f(-2)$) .

3° Compare $f(2)$ and $f(3)$.

In what follows we take $a = 5$, that is $f(x) = -x + 5 - \dfrac{4}{x}$ and we designate by (C) its representative curve in an orthonormal system.

4° Verify that the line (d) of equation $y = -x + 5$ is an asymptote to (C) .

5° Calculate $f(-2)$ and $f(2)$.

6° Show that the representative curve (C) cuts the abscissas' axis at the points of abscissas 1 and 4 .

7° Plot line (d) and curve (C) .

8° Solve graphically the inequality $f(x) < 0$.

9° Use (C) to plot the curve representing the function g defined by $g(x) = -f(x)$.